The irrationality exponent of $\log(3/2) + \sqrt{2}$ is at most four

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Abstract

We extend the paired-embedding affine determinant argument to fixed nonintegral algebraic data. A column factor clears the additional denominators while preserving the original matrix and its selected minor. The resulting logarithmic cost is absorbed by the dimension and approximation-weight choices, before the matrix height grows. The specialization $a = 3/2$, $u = 1$, $v = \sqrt{2}$ gives an irrationality exponent at most four for $\log(3/2) + \sqrt{2}$. The extension uses the interpolation and analytic estimates of the preceding affine development. We distinguish the mathematical argument, the accompanying Lean declarations, and their verification status; no assertion of optimality or historical priority is made.

1 Statement and contribution

For an irrational real number x, its irrationality exponent $\mu(x)$ is the supremum of the positive real ν for which $0 < |x - p/q| < q^{-\nu}$ holds for infinitely many reduced fractions p/q with $q > 0$. Our result is stated first as a literal uniform bound.

Theorem 1.1. *For every real $\nu > 4$, there is an integer $Q \geq 2$ such that*

$$q^{-\nu} \leq \left|\log(3/2) + \sqrt{2} - \frac{p}{q}\right| \qquad (p \in \mathbb{Z},\ q \in \mathbb{N},\ q \geq Q). \tag{1}$$

Consequently $\log(3/2) + \sqrt{2}$ is irrational and $\mu(\log(3/2) + \sqrt{2}) \leq 4$.

The threshold precedes both p and q; no coprimality assumption is needed in (1). The theorem does not identify the exponent exactly, compute Q, or assert that all exponents equal to four are excluded.

The preceding affine paper [1] treated algebraic-integer data in a degree-four field with two compatible embeddings. Its geometric and analytic statements do not require integrality. This note replaces the arithmetic step, accounts for its cost in the same parameter construction, and specializes the result to a rational base that was excluded by the old arithmetic hypothesis. The underlying interpolation and determinant estimates are inherited from the OpenAI library, the rational- and quadratic-logarithm extensions, and their affine adaptation [2, 1]; they are not new results of this note.

One checked published comparison comes from Marcovecchio [3], whose nonquadraticity table gives $\mu_2(\log(3/2)) < 9.460812$. With that paper's minimal-polynomial height, this implies the rational-exponent bound 18.921624 for our shifted target; Section 5.1 gives the conversion. This is an applicable benchmark, not a certification of the best published bound for this exact number.

2 The denominator-aware affine theorem

Write $\mathrm{Emb}(F) = \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(F, \mathbb{C})$. The ring of integers is denoted by $\mathcal{O}_F$.

Theorem 2.1. *Let $[F : \mathbb{Q}] = 4$ and $a, u, v \in F$. Suppose an embedding σ_0 satisfies $\sigma_0(a) = b_0 \in \mathbb{R}$, with $b_0 > 0$ and $b_0 \neq 1$. Let $x \in \mathbb{R} \setminus \{0\}$. Suppose a selected set $G \subset \mathrm{Emb}(F)$ has cardinality two and*

$$\exp(\sigma(u)(x - \sigma(v))) = \sigma(a), \qquad |\sigma(u)| \leq 2 \quad (\sigma \in G). \tag{2}$$

Then, for every real $\nu > 4$, there is $Q \in \mathbb{N}$, $Q \geq 2$, such that $q^{-\nu} \leq |x - p/q|$ for every $p \in \mathbb{Z}$ and $q \in \mathbb{N}$, $q \geq Q$. In particular, x is irrational and $\mu(x) \leq 4$.

Other embeddings may also satisfy (2), and σ_0 need not belong to G. Embeddings are counted as distinct maps even when their values on a, u, v coincide. The result does not require the remaining two embeddings to be real.

Choose positive integers d_a, d_n such that

$$d_a a, \quad d_n u, \quad d_n uv \ \in \mathcal{O}_F. \tag{3}$$

Every element of a number field has a positive integer multiple in its ring of integers. Taking a product of denominators clears the last two elements simultaneously. These choices depend only on the fixed algebraic data, not on any approximating fractions or matrix height.

2.1 The inherited analytic and geometric input

We state the precise part of the existing argument that is reused. For a fixed $\nu > 4$, suppose there are approximations $|x - p/q| < q^{-\nu}$ with arbitrarily large q. Choose finitely many p_i/q_i with $q_i \geq 2$ and $p_i \neq 0$, and put $w_i = \lceil \log q_i \rceil$, $w_* = \min_i w_i > 0$. Nonzero numerators are available by increasing the initial threshold, since $x \neq 0$. There are rational shape parameters

$$0 < \theta < A < B < 1, \quad C > 1, \quad B < C^{-1}, \quad B < C\theta < 1, \tag{4}$$

and $0 < \eta < 1$, for which

$$g = \nu(A(1 - \eta) - \theta) - 2(1 - \theta) > 0. \tag{5}$$

Choose $F_0 > 2/\theta$, a dimension $m \geq 1$, and set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{6}$$

The finitely many weights w_i are then chosen successively large enough to satisfy the weight-separation hypotheses of weighted logarithmic interpolation in [1, Sections 4–5]. These hypotheses compare finitely many products of earlier weights; an arbitrarily large next denominator suffices. The strict volume ratio is $Kw_0\theta^m/v_0 = 1/2$.

For $H > 0$, use columns $(h, \gamma) \in \mathbb{N} \times \mathbb{N}^m$ with $w_0 h + \sum_{i=1}^m w_i \gamma_i \leq H$, and rows (j, s, β) with $0 \leq j < K$ and $v_0 s + \sum_{i=1}^m w_i \beta_i/\theta < H$. The number of rows is $M = M(H) > 0$. Define

$$T_i = \lceil F_0 w_i / v_0 \rceil, \qquad L_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1}}{k} t^k, \tag{7}$$

$$r_{\sigma i} = \sigma(u)(p_i/q_i - \sigma(v)), \qquad z_\sigma = \sigma(a).$$

The matrix is

$$\mathcal{M}_\sigma(H)_{(j,s,\beta),(h,\gamma)} = z_\sigma^{jh} [t^s Z^\beta](1 + t)^h \prod_{i=1}^m (Z_i + j r_{\sigma i} + L_i(t))^{\gamma_i}. \tag{8}$$

Weighted interpolation gives full row rank at σ_0 for a cofinal set of heights. Choose one nonzero full-row minor there and use its *same columns* at every embedding; its determinant is Δ_σ. Write

$$b_H = \frac{1}{MH} \sum_{\text{rows } r} \sum_i w_i \beta_{ri}, \qquad 0 \le b_H \le \theta.$$

Fix bounds $\kappa, R \ge 0$ and $\rho \ge 100$ such that, at all embeddings,

$$\begin{aligned} \log \max(1, |\sigma(a)|) &\le \kappa, \\ |\sigma(u)|(|x| + 1 + |\sigma(v)|) &\le R, \\ 100|x| \le \rho, \qquad & 100|\sigma(u)(x - \sigma(v))| \le \rho. \end{aligned}$$

Finite sums or maxima provide such bounds before the approximations are selected. Put $\Lambda = \log 4 + 4$ and $c = \log 2/4$, and define

$$\begin{aligned} E_0 &= \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*}, \qquad (9) \\ U(S) &= \frac{\log 2}{v_0} + \frac{\kappa K + \log(3/2)}{w_0} + \frac{\log(2(KS+2))}{w_*}, \\ V &= \frac{\nu}{F_0} + \frac{2\log 2}{v_0} + \frac{\rho K}{w_0} + \frac{\log 4 + \log(2K) + \nu + \log(200K) + \log(\rho/100)}{w_*}. \end{aligned}$$

The inherited estimates [1, Sections 7–8] assert that there are remainders $r_a(H), r_c(H) \to 0$ and a collision rate $C_H \to C_\infty$, where

$$C_\infty = \frac{c\eta^2 K \theta^m}{(m+1) v_0 A^m} = \frac{c\eta^2}{2(m+1)} (B/A)^m, \qquad (10)$$

such that every nonzero selected minor satisfies

$$\begin{aligned} \frac{\log |\Delta_\sigma|}{MH} &\le V + r_a(H) + \frac{\log 2}{w_*} + \max\{-C_H, -\nu(A(1-\eta) - b_H)\}, \qquad \sigma \in G, \qquad (11) \\ \frac{\log |\Delta_\sigma|}{MH} &\le U(R) + r_c(H), \qquad \sigma \notin G. \qquad (12) \end{aligned}$$

These are estimates for the original matrix (8); no clearing factors are built into them. Their hypotheses are bounds on complex bases and centres, the exponential identities, the approximation errors and the numerical parameters. Algebraic integrality is absent. Likewise, interpolation permits arbitrary additive centres. This is why these statements remain applicable to the present data.

3 Clearing denominators without changing the minor

Let $D_i = \operatorname{lcm}(1, \dots, T_i)$ and $e_i = \lfloor H/w_i \rfloor$. The original row and column factors are

$$\prod_i D_i^{e_i} q_i^{-\beta_i} \quad \text{and} \quad \prod_i q_i^{\gamma_i}.$$

Their product over the selected square matrix is a positive rational number $\mathscr{C}_H$. The elementary clearing-cost calculation, which is an inequality about these numerical factors, gives

$$\log \mathscr{C}_H \le MH(1 - b_H + E_0). \qquad (13)$$

It does not itself assert integrality for the present data.

Add to each selected column (h,γ) the integer factor

$$c(h,\gamma) = d_a^{Kh} d_n^{\sum_i \gamma_i}, \tag{14}$$

and let A_H be the product of these extra factors over the selected columns. The exponent Kh is a convenient upper bound; replacing it by $(K-1)h$ is possible but unnecessary.

Lemma 3.1. *There is a single nonzero algebraic integer $Z_H \in \mathcal{O}_F$ such that*

$$\sigma(Z_H) = \mathscr{C}_H A_H \Delta_\sigma \qquad (\sigma \in \mathrm{Emb}(F)). \tag{15}$$

In particular, all four selected determinants are nonzero.

Proof. Put $n_i = u(p_i - q_i v)$ and $N_i = d_n n_i$. Equation (3) makes $N_i = (d_n u)p_i - (d_n uv)q_i$ integral. If some $\beta_i > \gamma_i$, the matrix entry is zero. Otherwise put $\delta_i = \gamma_i - \beta_i$. After extracting Z^β and applying the original factors, the entry is the embedded image of

$$a^{jh} \prod_i \binom{\gamma_i}{\beta_i} [t^s](1+t)^h \prod_i D_i^{e_i} (jn_i + q_i L_i(t))^{\delta_i}.$$

The additional factor (14) clears the remaining denominators because

$$d_a^{Kh} a^{jh} = d_a^{(K-j)h} (d_a a)^{jh},$$
$$d_n^{\gamma_i} (jn_i + q_i L_i)^{\delta_i} = d_n^{\beta_i} (jN_i + d_n q_i L_i)^{\delta_i}.$$

All exponents are nonnegative. The polynomial $D_i(jN_i + d_n q_i L_i)$ is integral coefficientwise, and $\delta_i \le \gamma_i \le e_i$, so its δ_ith power is cleared by $D_i^{e_i}$. Products, binomial coefficients and coefficient extraction preserve integrality.

Thus these entries define one square matrix over $\mathcal{O}_F$. Its determinant is Z_H, and determinants commute with every embedding. This proves (15). At σ_0 its image is nonzero, since both factors are positive and the selected minor is nonzero. Hence $Z_H \neq 0$, and injectivity of every field embedding proves the last assertion. No new column choice is made at the other embeddings. □

Lemma 3.2. *With*

$$\tau = \frac{K \log d_a}{w_0} + \frac{\log d_n}{w_*}, \tag{16}$$

one has $0 \le \log A_H \le MH\tau$.

Proof. Every column satisfies $h \le H/w_0$ and $\sum_i \gamma_i \le H/w_*$. Apply these inequalities to the logarithm of (14), then sum over M columns. The logarithms of the positive integer denominators are nonnegative. The transverse cost has no additional factor m, because the column budget bounds the sum of all exponents. □

The nonzero integer norm of Z_H gives

$$0 \le 4 \log(\mathscr{C}_H A_H) + \sum_\sigma \log|\Delta_\sigma|.$$

Using (13), Lemma 3.2, and the estimates at two good and two remaining embeddings, then dividing by $2MH > 0$, gives

$$-2(1-b_H) - 2(E_0+\tau) - U(R) - r_c(H) \le V + r_a(H) + \frac{\log 2}{w_*} + \max\{-C_H, -\nu(A(1-\eta) - b_H)\}. \tag{17}$$

The added normalized arithmetic cost is at most 2τ. It is counted only here; the analytic bounds still concern the unscaled determinants.

4 Absorption and the contradiction

Set

$$E = E_0 + \frac{K \log d_a}{w_0}, \qquad \widehat{R} = 2d_n^2(R+2).$$

For $K \geq 1$, $R \geq 0$, and $d_n \geq 1$,

$$2(K\widehat{R}+2) \geq 2d_n^2\, 2(KR+2).$$

Indeed the difference is $8d_n^2(K-1)+4 \geq 0$. Taking logarithms and dividing by $w_* > 0$ proves

$$U(R) + \frac{\log 2 + 2\log d_n}{w_*} \leq U(\widehat{R}). \tag{18}$$

This enlargement pays the transverse denominator cost and the good-embedding multiplier simultaneously.

We now justify the required parameter choices with these fixed costs. First choose shape parameters with (5); for example take rational $t < 1$ close enough to one that $\nu t > 2(1+t)$, put $\theta = t^2$, choose rational $\theta < A < B < t$ close to t, then choose $C \in (B/\theta, 1/B)$ and a sufficiently small $\eta > 0$. Choose $0 < \epsilon < g$ with $\epsilon \leq 1/2$, then choose $F_0 > 2/\theta$ with $\nu/F_0 < \epsilon/3$.

Choose m sufficiently large that

$$\frac{2\Lambda F_0 m + 3\log 2}{v_0} + \frac{(\rho+\kappa+2\log d_a)K + \log(3/2)}{w_0} < \epsilon/3, \qquad C_\infty > 3. \tag{19}$$

This is possible since $K/w_0 \leq (CB)^m \to 0$, $m/v_0 \to 0$ because $C\theta/B > 1$, and $C_\infty \to \infty$ by (10). Next choose the geometric slack required for interpolation and increase the initial approximation-weight threshold so that all remaining inverse-weight terms in $2E + U(\widehat{R}) + V$ sum to less than $\epsilon/3$. Their coefficients are fixed at this point, and $\sum_i 1/w_i \leq m/w_*$. Later denominators can still be chosen successively large enough for separation. Consequently

$$2E + U(\widehat{R}) + V < g, \qquad 2 + 2E + U(\widehat{R}) + V < C_\infty. \tag{20}$$

All these choices, including the finite approximation tuple, precede the increase of H. In particular $K\log d_a/w_0$ and $\log d_n/w_*$ are not height remainders.

By (18), (17) implies

$$-2(1-b_H) \leq S_H + \max\{-C_H, -\nu(A(1-\eta)-b_H)\},$$

where $S_H = 2E + U(\widehat{R}) + V + r_a(H) + r_c(H)$. For all sufficiently large H, (20) yields $S_H < g$ and $2 + S_H < C_H$. If the maximum is the approximation term, rearranging gives

$$g + (\nu-2)(\theta - b_H) = \nu(A(1-\eta)-b_H) - 2(1-b_H) \leq S_H < g,$$

a contradiction. If it is the collision term, it gives $C_H \leq 2(1-b_H) + S_H \leq 2 + S_H$, again a contradiction. Either argument works if the two terms are equal.

Interpolation supplies a nonzero minor at a cofinal set of heights, so this contradiction disproves the existence of arbitrarily large bad denominators. This proves the literal bound in Theorem 2.1. A rational x would have exact representations with arbitrarily large unreduced denominators, contrary to that bound. Thus x is irrational. For each $\nu > 4$, only finitely many reduced fractions can violate it: denominators are bounded, and the inequality bounds their numerators as well. Hence $\mu(x) \leq 4$.

5 Specialization and published comparison

Take $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which has degree four, and put

$$a = 3/2, \qquad u = 1, \qquad v = \sqrt{2}, \qquad x = \log(3/2) + \sqrt{2} > 0.$$

The two embeddings fixing $\sqrt{2}$ and sending $\sqrt{3}$ to its two signs are distinct. They satisfy (2), since $\exp(x - \sqrt{2}) = 3/2$, and their slopes have absolute value one. The positive distinguished base is $3/2 \neq 1$. We may choose $d_a = 2$, $d_n = 1$ in (3). Theorem 2.1 proves Theorem 1.1.

5.1 Height conversion

For coprime p, q, $q > 0$, the quadratic number $U = p/q - \sqrt{2}$ has primitive irreducible integer polynomial

$$q^2X^2 - 2pqX + (p^2 - 2q^2).$$

Primitivity follows from $\gcd(q^2, p^2 - 2q^2) = 1$; irreducibility follows from $\sqrt{2} \notin \mathbb{Q}$. Its coefficient height obeys $H(U) \geq q^2$. For approximations with $|x - p/q| < 1$, it also obeys $H(U) \leq C_x q^2$, where $C_x > 0$ is fixed. Marcovecchio's definition and table [3, pp. 147, 151, 182] imply, for every $s > 9.460812$ and all sufficiently large such heights,

$$|x - p/q| = |\log(3/2) - U| > H(U)^{-s} \geq C_x^{-s} q^{-2s}.$$

Choosing $9.460812 < s < \nu/2$ and absorbing the fixed constant gives the comparison bound $\mu(x) \leq 18.921624$. The factor two comes from the quadratic growth of coefficient height. The comparison checks an applicable consequence of that result; it does not settle the strongest current bound or priority for four.

6 Formalization and verification status

The new source is maintained separately from the inherited affine package. Its main mathematical boundary is the entry lift and the positive scalar for the same selected determinant. The numerical parameter structures are reused with base-denominator parameter $\log d_a$ and coarse bound $\widehat{R}$.

Source validation. The general denominator-aware endpoint and the pilot corollaries have compiled in Lean 4.34.1 with strict warning settings. The ten new source files are identified by a SHA-256 inventory. A transitive axiom-policy audit passed for all seven public endpoints, with rejection controls. Separate mathematical, specification and blind manuscript reviews found no remaining mathematical or statement defect in this scope.

Independent verification. Six statements passed comparison against a separate challenge and both of Comparator's kernel checks. Nanoda and a separately built stock Lean 4.34.1 reader then accepted identical retained export bytes. The export contains 112854 serialized declarations, including proof bodies for all six public targets and all six comparison wrappers, and exactly the three permitted axioms. Source and export hashes, acceptance and rejection controls, and tool identities accompany the package. Local execution used identified inherited caches; a fresh hosted build and its evidence audit remain pending. The pinned Nanoda checker accepts a known invalid-projection test that both stock Lean checks reject. Its result is supplementary; the stock Lean replay remains required.

The standard-definition corollary excludes `LiouvilleWith`(ν, x) for every $\nu > 4$. To absorb the constant in that definition, apply the literal bound at $\nu' = (4 + \nu)/2$ and use the frequently occurring smaller-exponent approximations furnished by Mathlib. Irrationality is proved separately, so the statement does not rest on a default supremum value.

Attribution and limitations

This work reuses the interpolation, collision, logarithmic-jet and parameter-selection machinery in the OpenAI source and the preceding affine adaptation [2, 1]. The new step is the fixed-denominator arithmetic extension and its specialization above. Code provenance, licenses and exact declaration correspondence belong to the accompanying formalization package. The present paper does not claim a formalization of every possible quadratic-field affine family, an effective approximation threshold, or a new best published record.

OpenAI Codex assisted with mathematical development, Lean proof construction, verification orchestration and exposition. Separate model contexts perform mathematical, specification and manuscript reviews. Such reviews are not human peer review, and kernel acceptance does not establish historical priority.

References

[1] R. M. Casper, *The irrationality exponents of* $\log 2 + \sqrt{2}$ *and* $\sqrt{2} \log 2$ *are at most four*, October 11, 2026. Companion preprint and `AffineLogarithms` formalization in the `Mattie/math` repository.

[2] OpenAI, *`OAI.NumberTheory.PiExponent`*, Lean source library, revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, https://github.com/openai/math. The rational-logarithm extensions are inherited through the `Mattie/math` revision `0cfe10002ea95c45721b93d18bffbe2c3bbbcbbd`; the `AffineLogarithms` package records their exact provenance.

[3] R. Marcovecchio, *The Rhin–Viola method for* $\log 2$, Acta Arithmetica **139** (2009), no. 2, 147–184. doi:10.4064/aa139-2-5.